

Relative Valuation

What is the purpose of using multiples in valuation?

A valuation multiple is a financial ratio that reflects how valuable a particular company is in relation to a specific metric. The numerator will be a measure of value such as equity value or enterprise value, whereas the denominator will be a financial or operating metric.

Since absolute values cannot be compared, multiples are used to standardize a company's value on a per-unit basis. This enables comparisons in value amongst similar companies, which is the premise of relative valuation. For any valuation multiple to be meaningful, a contextual understanding of the target company, such as its fundamental drivers and general industry knowledge, is required.



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Valuation Multiples

How do you determine what the appropriate numerator is for a multiple?

For multiples, the represented stakeholders in the numerator and denominator must match.

- If the numerator is enterprise value, metrics such as EBIT, EBITDA, unlevered free cash flow (FCFF), and revenue multiples can be used since these are all unlevered (i.e., pre-debt) measures of profitability.
- In contrast, if the numerator is equity value, metrics such as net income, levered free cash flow (FCFE), and earning per share (EPS) would be used because these are all levered (i.e., post-debt) measures.

Walk me through the process of “spreading comps.”

Before you can answer the question, you must ask for clarification on whether the interviewer is asking about trading or transaction comps. The processes for both, however, have many overlapping aspects.

1. **Determine Comparable Peer Group:** The first step to perform comps is to select the peer group. For trading comps, the peer group will be composed of publicly traded comparable companies that are competitors in the same industry or operate within a nearby industry. For transaction comps, the peer group would include companies recently involved in M&A deals within the same or a similar industry.
2. **Collect Relevant Information:** The next step involves finding publicly available information that may be helpful to understand the trends and factors affecting how companies in a particular industry are being (or were) valued. Most of the insights gathered in this step will be more on the qualitative side and related to industry research, understanding ongoing developments, and company-specific details.
3. **Input Financials:** With the industry research completed, you'll then pull the financial data of each comparable company and then “scrub” the financials for non-recurring items, accounting differences, financial leverage differences, and business life cycles (cyclicality, seasonality) to ensure consistency and allow for a fair comparison among the companies. If relevant, you'll also calendarize each peer group company's financials to standardize the metrics to ensure comparability.
4. **Multiples Calculation:** Then, the peer group's valuation multiples will be calculated and benchmarked in the output sheet. At a minimum, the multiples are shown on a last twelve months (LTM) and the next fiscal year (NTM) basis, and as a general convention, the minimum, maximum, 25th percentile, 75th percentile, mean and median will be listed. Using the research collected in previous steps, you'll then attempt to understand the factors causing the differences and remove any outliers if deemed appropriate.
5. **Apply Multiple to Target:** In the last step, the target company being valued will have the median (or mean) multiple applied to the corresponding metric to arrive at its approximate comps-derived value. Understanding the fundamental drivers used to value companies within a particular industry makes comps-derived valuations defensible – otherwise, justifying whether the target should be valued on the higher or lower end of the valuation range will be difficult.

When putting together a peer group for comps, what would some key considerations be?

- **Operational Profile:** These characteristics entail the nature of the business, such as its industry, business model, products/services sold, and end markets served. In addition, the company's position within the market (market leader or market challenger), stage in the life cycle, and seasonality/cyclicality should all be considered. The importance of selecting the right peer group for a comps-derived valuation cannot be overstated, and this begins with understanding the target's operational characteristics.
- **Financial Profile:** If the company might be a suitable inclusion to the peer group based on its operational characteristics, its financial profile would then be considered. Some metrics to gather would be the company's key cash flow metrics, size in terms of valuation, profitability margins, credit ratios, historical/estimated growth rates, and return metrics.

Operational Profile	Financial Profile
Industry/Sector	Size (Enterprise Value, Equity Value)
Product and Service Offerings	Margin Profile (Gross Margin, Operating Margin, EBITDA Margin)
End Markets and Customers Type	Historical and Forecasted Growth Rates (Revenue, EBITDA, EPS)
Distribution Channels	Credit Profile (Leverage Ratios, Coverage Ratios)
Cyclicality or Seasonality	Return Metrics (ROIC, ROA, ROE)

Should the target company being valued be included in its peer group?

Many professionals exclude the target company being valued from the peer group because the target's inclusion would skew the multiple towards the target's current valuation. However, if the intuition behind a comps analysis is that the market may misprice individual stocks but is correct on the whole, then logic dictates that the target should be included in its market-based valuation.

What are the primary advantages of the trading comps approach?

- **Public Filings:** Trading comps involves public companies, making data collection far more convenient since all their reports and filings are easily accessible online.
- **Less Data Required:** Implementation is a key advantage of trading comps, as proper DCFs cannot be built without detailed financials and supplementary data. But to get a decent trading comps-based valuation, only a few data points (e.g., EBITDA, revenue, net income) are required, making it easier to value companies when access to data sets is limited.
- **Current Valuations:** Trading comps reflect up-to-date, current valuations based on investor sentiment as of the present day, since it's based on the latest prices paid in the public markets.

What are the main disadvantages of performing trading comps?

- Putting together a peer group of "pure-play" companies by itself can be a challenging task, especially if the target is differentiated and has few (or no) direct competitors.
- Even with a well-thought-out, similar peer group, explaining the differences in valuation can be difficult as the comparison is always "apples to oranges."
- Understanding valuation gaps between a company and its comparables involves judgment, which can be very challenging – plus, the market is often emotional and fluctuates irrationally, bringing in more external factors to consider.



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Trading Comps
Strengths and
Weaknesses**

- Low trade volume and less followed equities may not reflect their true fundamental value, making them less useful for trading comps.

To perform transaction comps, how would you compile the data?

When collecting the data to perform transaction comps, you would use deal announcement press releases, proxy filings, and the merger agreement to learn about the deal terms. You would also use the target company's filings (annual and quarterly reports) for historical financial data, research reports, and financial data vendors such as Bloomberg, Capital IQ, or FactSet for historical share price data and estimated earnings forecasts.

What do transaction comps tell you that trading comps cannot?

Transaction comps can provide insights into control premiums that buyers and sellers should expect when negotiating a transaction.

In addition, transaction comps can validate potential buyers' existence in the private markets and if a particular investment strategy has been successfully implemented before.

Let's say a certain company is valued at a specific price based on a DCF analysis and confirmed to be within range by trading comps. However, if there are no buyers in the market, the seller is unlikely to exit at its expected valuation.



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Transaction Comps**

In M&A, why is a control premium paid?

A control premium refers to the amount an acquirer paid over the market trading value of the shares being acquired (usually shown as a percentage). As a practical matter, a control premium is necessary to incentivize existing shareholders to sell their shares. It's improbable that an acquirer could get a controlling interest in a target company without first offering a reasonable purpose premium over the current price.

From the perspective of the shareholders of the acquisition target: "What would compel existing shareholders to give up their ownership if doing so is not profitable?"

Besides incentivizing existing shareholders to sell, what other factors lead to higher control premiums being paid?

- **Competitive Deals:** In M&A, nearly all acquirers pay a control premium due to the competitive elements of sale processes. As a general rule, the control premium paid will be higher the more buyers are involved, as competition directly drives up the price.
- **Synergies:** If there are potential synergies that can be realized by the acquirer and the management team has high conviction in its occurrence, then the acquirer might be justified in paying a higher premium.
- **Asset Scarcity:** Many acquirers (strategics in the majority of cases) might pay a higher premium if the specific asset is a centerpiece to their future plans and there are no other acquisition targets in the market that meet their criteria. Oftentimes, this acquisition could lead to a meaningful competitive advantage for the acquirer over the rest of the market, making the completion of this acquisition a necessity.
- **Undervalued Target:** The target company might be perceived to be significantly undervalued from the buyer's viewpoint. From their perspective, the purchase price could be a moderate premium when compared to their own fair value assessment of the target, whereas to others the buyer paid an unreasonably high premium.
- **Mismanagement:** A mismanaged company coincides with the previous point, as this typically leads to underperformance. The acquirer will most likely be under the impression that the management should be replaced, and through operational improvements, a significant amount of value that is currently not being fully taken advantage of could be derived. Thus, a target acquired for this reason will be immediately restructured, beginning with the management team being replaced post-closing.

Why is transaction comps analysis often more challenging than trading comps?

Performing comparable acquisition analysis can be challenging when there has been limited (or no) M&A activity within the relevant space, or the comparable transactions were completed a long time ago in a completely different economic environment. Data from those prices paid might not reflect current market trends and investor sentiment.

The transaction dates place a significant constraint on the pool of comparable transactions. Generally, only relatively recent transactions (within the last five years) offer insight into industry valuations.

The transaction context must also be looked at, such as the form of consideration, the type of buyer (financial or strategic), and the deal's circumstances, which can significantly influence the final purchase price. However, this information about the deal can be challenging to compile, especially when they involve private companies, as they're not required to disclose all this deal-related information. Most times, even the purchase price paid for a company may not even be announced – thus, the data found is “spotty” and less straight-forward than trading comps.



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Transaction Comps
Strengths and
Weaknesses

When putting together a peer group for transaction comps, what questions would you ask?

- What was the transaction rationale from both the buyer and seller's perspective?
- Was the acquirer a strategic or a financial buyer?
- How competitive was the sale process?
- Was the transaction an auction process or negotiated sale?
- What were the economic conditions at the time of the deal?
- Was the transaction hostile or friendly?
- What was the purchase consideration?
- If the industry is cyclical (or seasonal), did the transaction close at a high or low point in the cycle?

When valuing a company using multiples, what are the trade-offs of using LTM vs. forward multiples?

Using historical (LTM) profits have the advantage of being actual results. This is important because EBITDA, EBIT, and EPS forecasts are subjective and especially problematic for smaller public firms, whose guidance is less reliable and harder to obtain.

That said, LTM suffers from the problem that historical results are often distorted by non-recurring expenses and income, misrepresenting the company's recurring operating performance. When using LTM results, non-recurring items must be excluded to get a “clean” multiple. In addition, companies are often acquired based on their future potential, making forward multiples more relevant.

Therefore, both LTM and forward multiples are often presented side-by-side, rather than picking one.

Read More → [The Role of Multiples in Valuation](#)

Why might two companies with identical growth and cost of capital trade at different P/E multiples?

Growth and cost of capital are not the only drivers of value. Another critical component is the return on invested capital (ROIC). Besides having different ROICs, the two companies could very well just be in different industries or geographies.

Other reasons may include relative mispricing or inconsistent EPS calculations, often caused by non-recurring items or different accounting policies.

Should two identical companies with different leverage ratios trade at different EV/EBITDA multiples?

You would expect the EV/EBITDA multiples to be similar because enterprise value and EBITDA measure a company's value and profits independent of its capital structure. Technically, they won't be exactly equal because enterprise value depends on the cost of capital, so there'll be some variation.

Should two identical companies with different leverage ratios trade at different P/E multiples?

P/E multiples can vary significantly due to leverage differences for otherwise identical companies. All else being equal, as a company borrows debt, the EPS (denominator) will decline due to higher interest expense.

The impact on the share price, on the other hand, is hard to predict and depends on how the debt is used:

- If the debt proceeds go unused and generate no return, the share price will decline to reflect the incremental cost of debt with no commensurate growth or investment. In this scenario, the share price and P/E ratio can be expected to decline.
- But if that debt were used to invest and grow the business, the P/E ratio should increase.

Simply put, debt adds more risk to equity investors (given their junior position in the capital structure) with little potential return, and investors will value the company at a lower P/E.

Which multiples are the most popular in valuation?

Enterprise Value/EBITDA multiples are the most common, followed by EV/EBIT and P/E. There are several others that are more industry and company-specific. For example, P/B ratios are used to value financial institutions, EV/Revenue multiples are used to value unprofitable companies, and (EV – Capex)/EBITDA multiples can be used for capital-intensive industries such as manufacturing or cable companies.

What are some common enterprise and equity value multiples?

- **Enterprise Value Multiples:** EV/Revenue, EV/EBIT, EV/EBITDA
- **Equity Value Multiples:** Price/Earnings (P/E), Price/Book (P/B), Price/Levered Cash Flows

Read More → [What Does an EV/EBITDA Multiple Mean?](#)

Why would it be incorrect to use enterprise value and net income in a multiple?

There would be a mismatch between the represented investor groups in the numerator and denominator. Enterprise value represents the value of the operations to all stakeholders in a company, meaning the cash flows belonging to both lenders and equity providers of capital. Net income, however, represents the residual value that flows just to equity shareholders.

Why might one company trade at a higher multiple than another?

One company may be valued at a higher multiple than another because of superior fundamentals such as better growth prospects, higher return on invested capital, lower cost of capital (WACC), and more robust cash flow generation. Investors are forward-looking. Therefore, companies with better growth trajectories and signs of being well managed with efficient capital allocation are rewarded with higher valuations.

Intuitively, what does the P/E ratio mean?

The price-to-earnings ratio (P/E) is one of the most widely used metrics by investors to determine a company's relative value against the industry average and its peers. This can then help determine whether the company is undervalued, fairly valued, or overvalued. The P/E ratio answers: "How much is the market willing to pay for a dollar of this company's earnings?"

$$\text{PE Ratio} = \frac{\text{Current Share Price}}{\text{EPS}}$$

$$\text{PE Ratio} = \frac{\text{Market Capitalization}}{\text{Net Income}}$$

A company is currently trading at 12.5x EBITDA, is it overvalued?

The answer depends on the company's industry being valued and how its valuation fares compared to the rest of its peer group. For instance, a software company trading at 12.5x would be near the sector median, whereas a double-digit multiple in an industry such as Oil & Gas is rare.

We recommend researching various industries and identifying those that consistently receive higher valuations on average (e.g., software, healthcare) and which ones don't – and then figure out the reason. The differences can often be related to cyclical trends that are impacting an industry. For example, sub-industries related to cloud computing, cybersecurity, remote healthcare, gaming, and video streaming services have seen significant valuation increases since the beginning of the pandemic.

Read More → [Enterprise Value Multiples by Sector](#)

What does a high P/E ratio relative to a peer group imply?

If a company has a high P/E ratio in relation to its peer group, this might mean the market is overvaluing the company. Alternatively, it could mean the company has less earnings than its peer group (or is a younger company). For this reason, the use of P/E multiples is most useful when the company is mature and its peer group has a similar capital structure.

A high P/E ratio is almost always a positive sign that the market expects the company's earnings to grow in the future and is optimistic about its growth prospects, as well as its current capex (which is causing the lower earnings) will pay off over time and eventually lead to greater profits.

A company has an EPS of \$2.00 that has declined to \$1.00 four years later. Assume its share price has remained the same at \$10. Is its current P/E ratio higher or lower than its four year-back P/E ratio, and how would you interpret this situation?

The company's P/E ratio would be higher. The company's four year-back P/E ratio was 5.0x and its current P/E ratio is 10.0x. Therefore, its P/E doubled after its EPS declined by \$1.00.

One potential interpretation is that the company is now overvalued and shouldn't be trading at a 10.0x multiple given the deteriorated EPS. Realistically, the market would view a decrease in EPS negatively, resulting in a lower share price and valuation.

But since the share price didn't change, there are a few possible explanations:

1. The company could have issued more shares to raise capital that had a dilutive impact on EPS.
2. The company might have made a dilutive acquisition with stock as the main purchase consideration.

In both cases, the cause of the lower EPS is the increase in the denominator. The share price remaining constant suggests the market reaction to how the company plans to use the newly raised capital or the M&A deal was positive, otherwise, the share price would've dropped. Similar to how EPS can be artificially boosted from share buybacks, it can decrease without there being an actual drop in performance. The issuance of additional shares typically results in the per-share value decreasing from the dilution, but the \$10 share price is the post-dilution share price (meaning, the price was upheld despite the dilution).

When would the use of the PEG ratio be appropriate?

The PEG ratio is used to standardize the P/E ratio and enable comparisons among companies with different expected long-term growth rates (g). As a broad rule of thumb, a stock is fairly valued when the PEG ratio is 1, undervalued when the PEG ratio is below 1, and overvalued when above 1.

$$\text{PEG Ratio} = \frac{\left(\frac{P}{E}\right)}{g}$$

There are two approaches for the growth rate used:

1. **Trailing PEG:** In the first option, the growth rate can be based on a company's historical growth. For example, the LTM growth rate could be used, or the annualized growth rate over the past 3-5 years.
2. **Forward PEG:** The other, more commonly used option calculates the growth rate from a 1-3 year projection based on consensus estimates or an annualized version of those growth rates.

When would you value a business using the P/B ratio?

The P/B ratio can be used when the company's book value captures a substantial part of its real value. An example would be commercial banks, as most of their assets and liabilities are frequently re-valued and similar to their actual market values.

When should you value a company using a revenue multiple vs. EBITDA?

Companies with negative profits and EBITDA will have meaningless EBITDA multiples. As a result, multiples based on revenue are the only available option to gain some level of insight.

How would you value a pre-revenue early stage internet company?

An internet company with no revenue or negative profitability could be valued using user engagement metrics such as the subscriber count, monthly active users, daily active users, and website hits. For example, pre-revenue internet startups are often valued using multiples such as EV/DAU or EV/MAU.

Why is calendarization a required step when performing comps analysis?

Calendarization involves aligning the fiscal year ending dates of a group of comparable companies to allow for a more accurate comparison. The general convention is to adjust the financials so that all the fiscal years end in December. The reason calendarization is necessary is that cyclical and seasonality can skew results and create discrepancies when making comparisons.

If the market matters most when valuing public companies, do we even need a comps analysis?

Why not just use the market cap directly to value the company?

If the market were perfectly efficient, it would price individual equities correctly, rendering a comps analysis pointless. However, the reasoning for still using a comps analysis for public company valuations is that the market may be efficient on average, but it can be off when pricing individual companies. Most investors will agree that the market is relatively efficient. However, investors can be overly emotional and overreact, creating investment opportunities.

When applying a peer group derived EV/EBITDA multiple onto your target company, what is an argument for using the group's median multiple instead of the mean?

The simple reason is: Medians remove the distortive impact of outliers on the peer group multiple. When using larger peer groups, the median is preferable over the mean because it limits distortions from outliers, which increases as more companies are included. The mean might be preferable for smaller peer groups with fewer than five comparable companies and no outliers.